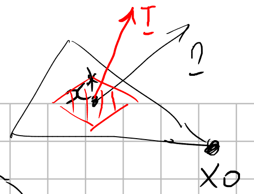
63.3



$$\left\{ x^*,\ \underline{n},\ \underline{\underline{\Gamma}}(\underline{x}) \right\} \rightarrow \underline{\underline{\Gamma}}(\underline{x}^*)$$

$$\begin{array}{c|ccc} & x & y & z \\ \hline x & 0 & 1 & 2 \\ y & 1 & 2 & 0 \\ z & 2 & 0 & 1 \end{array}$$

$$\underline{T}(\underline{x}^*) = \underline{\underline{\Gamma}}(\underline{x}^*)\cdot \underline{n}$$

x^{**}

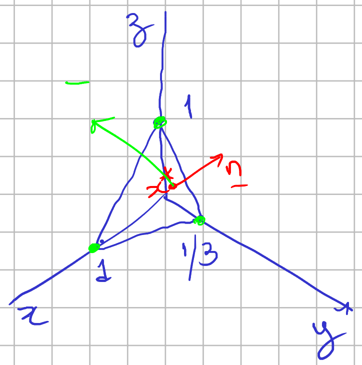
$$x+3y+z=1$$

$$n^P = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$

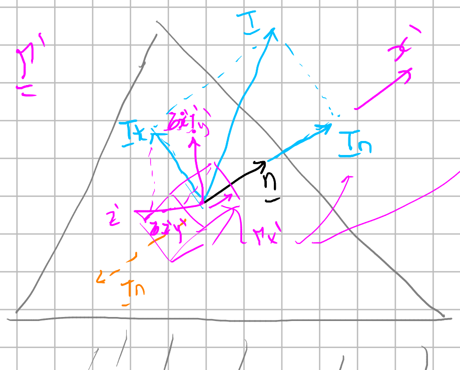
$$1.(\overbrace{x-0}^{x}) + 3(\underbrace{y-0}_{y=0}) + 1.(\overbrace{\underbrace{z-1}_{z=0}}^{0-1}) = 0 \qquad x_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

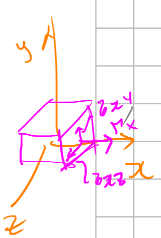
$$\underline{n}^P = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$



$$\underline{n} = \frac{\underline{n}^P}{\|\underline{n}^P\|} = \frac{\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}}{\sqrt{1^2+3^2+1^2}} = \frac{1}{\sqrt{11}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$$\underline{T} = \underline{\underline{\Gamma}}\cdot \underline{n} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix} \cdot \frac{1}{\sqrt{11}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{11}} \begin{bmatrix} 5 \\ 7 \\ 3 \end{bmatrix}$$





podria ser negativo

$$T_n = \underline{T}\cdot\underline{n} = \overbrace{\frac{1}{\sqrt{11}}\begin{bmatrix} 5 \\ 7 \\ 3 \end{bmatrix}}^{T} \cdot \overbrace{\frac{1}{\sqrt{11}}\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}}^{n} = \frac{29}{11}$$

$$\underline{T_n} = T_n \cdot \underline{n} = \frac{29}{11} \cdot \frac{1}{\sqrt{11}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \frac{29}{11\sqrt{11}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$$\|\underline{T_n}\| = |T_n|$$

$$\underline{T} = \underline{T_n} + \underline{T_t} \Rightarrow \underline{T_t} = \underline{T} - \underline{T_n} = \frac{1}{\sqrt{11}}\begin{bmatrix} 5 \\ 7 \\ 3 \end{bmatrix} - \frac{29}{11\sqrt{11}}\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \frac{1}{11\sqrt{11}}\begin{bmatrix} 26 \\ -10 \\ 4 \end{bmatrix}$$